A cycle embedding problem for Cube-Connected Cycles Graphs *

Yen-Hsun Ho, Chey-Woei Tsay, Lih-Hsing Hsu
Department of Computer Science and Information Engineering
Providence University, Taichung, Taiwan 43301, R.O.C.

Tung-Yang Ho†
Department of Television and Internet Marketing Management
Ta Hwa Institute of Technology, Hsinchu, Taiwan 30740, R.O.C.

Abstract

Let n be an odd integer with $n \geq 3$. The cube-connected cycles graph CCC_n has $n \times 2^n$ vertices, labeled $(l, \mathbf{x})$, where $0 \leq l \leq n-1$ and $\mathbf{x}$ is an n-bit binary string. Two vertices $(l, \mathbf{x})$ and $(l', \mathbf{y})$ are adjacent if and only if either $\mathbf{x} = \mathbf{y}$ and $|l - l'| = 1$, or $l = l'$ and $\mathbf{y} = (\mathbf{x})^l$. In this paper, we will find the set of all possible lengths of cycles in CCC_n with n is odd and $n \geq 3$.

Keywords: hamiltonian, cube-connected cycles network, vertex-fault-tolerant hamiltonian.

1 Introduction

For the graph definition and notation, we follow [6]. The cube-connected cycle CCC_n was introduced by Preparata and Vuillemin in 1981 [7] as a good alternate for the hypercube, having a fixed degree equal to 3 and yet a small diameter compared to its number of vertices. This graph was since then studied by a lot of people, who showed in particular, that it also has good properties as far as communications are concerned. It was proved that CCC_n contains a hamiltonian cycle, which is an interesting property to realize distributed algorithms. However, its cycle structure is still not completely known, as to whether it contains cycles of any given length. Germa, Heydemann, and Sotteau [4] further study the existence of cycles of all lengths in the cube-connected cycles graph and obtain the following theorem.

Theorem 1.1 *[4] The cube-connected cycles graph CCC_3 contains cycles of length 3 and all lengths between 8 and 24. CCC_4 contains cycles of length 4 and cycles of all even lengths between 8 and 64. For $n \geq 5$, CCC_n contains cycles of length n and of every even length between 8 and $n2^n$ except 10 and possibly $n2^n - 2$. Furthermore, for n odd, CCC_n contains cycles of every odd length between $n + 6$ and $n2^n - n - 2$ and also cycles of length $n2^n - n + 2$.*

Let $L(n)$ denote the set of all possible lengths of cycles in CCC_n. We restate Theorem 1.1 into the following theorem.

Theorem 1.2 *Let n be a positive integer with $n \geq 3$.*
(1) $L(3) \supseteq \{3\} \cup \{i \mid 8 \leq i \leq 24\}$;
(2) $L(4) \supseteq \{4\} \cup \{i \mid i$ is even and $8 \leq i \leq 64\}$;
(3) $L(n) \supseteq \{n\} \cup \{i \mid i$ is even and $8 \leq i \leq n+5\} \cup \{i \mid n+6 \leq i \leq n2^n\} - (\{10, n2^n - 2, n2^n - n\} \cup \{n2^n - j \mid j$ is odd and $1 \leq j \leq n-4\})$ if n is odd and $n \geq 5$; and
(4) $L(n) \supseteq \{n\} \cup \{i \mid i$ is even, $8 \leq i \leq n2^n$, and $i \neq 10, n2^n - 2\}$ if n is even and $n \geq 6$.

Recently, Ho et. al. [5] proved that $n2^n - 1 \in L(n)$ if n is odd and $n \geq 3$. In this paper, we

*This work was supported in part by the National Science Council of the Republic of China under Contract NSC 99-2115-M-233-001-MY2.
†Correspondence to: hoho@thit.edu.tw

compute $L(n)$ as follow:

Theorem 1.3 *Let n be an odd integer with $n \geq 3$. $L(n) = \{n\} \cup \{i \mid i$ is even, $8 \leq i \leq n+5$, and $i \neq 10\} \cup \{i \mid n+6 \leq i \leq n2^n\}$.*

2 Notation and definition

Let $\mathbf{u} = u_{n-1}u_{n-2}\ldots u_1u_0$ be an n-bit binary string. We use $(\mathbf{u})_j$ to denote the bit u_j. Let $\mathbf{0}^n$ denote the n-bit binary string with $(\mathbf{0}^n)_i = 0$ for $0 \leq i \leq n-1$. The *Hamming weight* of $\mathbf{u}$, denoted by $w(\mathbf{u})$, is the number of i such that $(\mathbf{u})_i = 1$. Let $\mathbf{u} = u_{n-1}u_{n-2}\ldots u_1u_0$ and $\mathbf{v} = v_{n-1}v_{n-2}\ldots v_1v_0$ be two n-bit binary strings. The *Hamming distance* $h(\mathbf{u}, \mathbf{v})$ between two vertices $\mathbf{u}$ and $\mathbf{v}$ is the number of different bits in the corresponding strings of both vertices. The *n-dimensional hypercube*, denoted by Q_n, consists of all n-bit binary strings as its vertices and the two vertices $\mathbf{u}$ and $\mathbf{v}$ are adjacent if and only if $h(\mathbf{u}, \mathbf{v}) = 1$. Obviously, Q_n is a bipartite graph with bipartition $A = \{\mathbf{u} \mid w(\mathbf{u})$ is even$\}$ and $B = \{\mathbf{u} \mid w(\mathbf{u})$ is odd$\}$.

The cube-connected cycles graph CCC_n has $n \times 2^n$ vertices, labeled $(l, \mathbf{x})$, where l is an integer between 0 and $n-1$, called the *level* of the vertex, and $\mathbf{x}$ is an n-bit binary string. Two vertices $(l, \mathbf{x})$ and $(l', \mathbf{y})$ are adjacent if and only if either $\mathbf{x} = \mathbf{y}$ and $|l - l'| = 1$ or $l = l'$ and $\mathbf{y} = (\mathbf{x})^l$. In the latter case, $\mathbf{x}$ and $\mathbf{y}$ only differ in the bit at position l. The edges joining two vertices of the same level are called *cube-edges*. To emphasize the level, the edge joining two vertices of level i is an *i-dimensional edge*. The edges that connect $(l, \mathbf{x})$ to its neighbors $(l+1, \mathbf{x})$ and $(l-1, \mathbf{x})$ are called *cycle-edges*. Moreover, these cycle-edges form a cycle of length n called the *fundamental cycle* defined by $\mathbf{x}$. For $n = 2$, CCC_n is simply the cycle of length 8. Therefore, in this article we will only consider the case $n \geq 3$.

Let $a \in \{0, 1\}$. We set CCC_n^a to denote the subgraph of CCC_n induced by the vertex set $\{(l, \mathbf{x}) \mid 0 \leq l \leq n-1$ and $(\mathbf{x})_{n-1} = a\}$. Obviously, $\deg_{CCC_n^a}((n-1, \mathbf{x})) = 2$ for any $\mathbf{x}$ with $(\mathbf{x})_{n-1} = a$. We replace the path $\langle (n-2, \mathbf{x}), (n-1, \mathbf{x}), (0, \mathbf{x}) \rangle$ with edge $((n-2, \mathbf{x}), (0, \mathbf{x}))$ for any $\mathbf{x}$ with $(\mathbf{x})_{n-1} = a$. The resultant graph is $(CCC_n^a)'$. It is easy to check that $(CCC_n^a)'$ is isomorphic to CCC_{n-1} for every $a \in \{0, 1\}$.

With the above observation, we recursively construct CCC_n from four copies of CCC_{n-2} through the following three steps.

Step 1: Let $ab \in \{00, 01, 10, 11\}$. We set $(CCC_n^{ab})'$ as the copy of CCC_{n-2} such that the vertex $(l, \mathbf{x})$ in $(CCC_n^{ab})'$ corresponds to the vertex $(l, \mathbf{y})$ in CCC_{n-2} satisfying $(\mathbf{x})_{n-1} = a$, $(\mathbf{x})_{n-2} = b$, and $(\mathbf{x})_i = (\mathbf{y})_i$ for $0 \leq i \leq n-3$. Note that $\mathbf{x}$ is an n-bit binary string and $\mathbf{y}$ is an $(n-2)$-bit binary string. In other words, two more bits ab are added to each vertex of CCC_{n-2}.

Step 2: For any n-bit binary string $\mathbf{x}$, insert all the $(n-2, \mathbf{x})$ and $(n-1, \mathbf{x})$ vertices. In other words, we replace the edge $((n-3, \mathbf{x}), (0, \mathbf{x}))$ with the path $\langle (n-3, \mathbf{x}), (n-2, \mathbf{x}), (n-1, \mathbf{x}), (0, \mathbf{x}) \rangle$. Obviously, the subgraph induced by those vertices with $(\mathbf{x})_{n-1} = a$ and $(\mathbf{x})_{n-2} = b$ is CCC_n^{ab}.

Step 3: Add all $(n-2)$-dimensional and $(n-1)$-dimensional edges to connected CCC_n^{00}, CCC_n^{01}, CCC_n^{10}, and CCC_n^{11}. The resultant graph is CCC_n.

With this observation, we introduce the following notation. For each subgraph G of CCC_{n-2}, we use G^{ab} to denote the corresponding subgraph graph in CCC_n^{ab}. We use $(G^{ab})'$ to denote the corresponding subgraph graph in $(CCC_n^{ab})'$. Then we replace the edge $((n-3, g(\mathbf{x})), (0, g(\mathbf{x})))$ with the path $\langle (n-3, g(\mathbf{x})), (n-2, g(\mathbf{x})), (n-1, g(\mathbf{x})), (0, g(\mathbf{x})) \rangle$ if the edge $((n-3, g(\mathbf{x})), (0, g(\mathbf{x})))$ is in F to obtain a subgraph of CCC_n^{ab}. The resultant graph is called G^{ab}.

3 A useful lemma

Let C be a cycle of CCC_n. The *subgraph of n-dimensional hypercube Q_n induced by C*, denoted by $Q_n(C)$, is the graph with $V(Q_n(C)) = \{\mathbf{x} \mid (l, \mathbf{x})$ is a vertex in C for some $l\}$ and $E(Q_n(C)) =$

$\{(\mathbf{x},\mathbf{y}) \mid ((l,\mathbf{x}),(l,\mathbf{y}))$ is an edge in C for some $l\}$. The *level set* of C, $LS(C)$, is the set $\{l \mid (l,\mathbf{x})$ is a vertex in $C\}$. We have the following lemma.

Lemma 3.1 *Let C be a cycle of CCC_n and $l(C)$ denote the length of cycle C. Then*
(1) $3 \le l(C) \le n2^n$.
(2) $Q_n(C)$ is either a graph with a single vertex, a four cycle, or a graph with at least 6 edges.
(3) $l(C) \ge 2|E(Q_n(C))|$.
(4) Let $l(C) \in \{3,5,7\}$. Then $l(C) \in L(n)$ if and only if $l(C) = n$.
(5) $9 \in L(n)$ if and only if $n \in \{3,9\}$.
(6) $10 \in L(n)$ if and only if $n \in \{3,4,10\}$.
(7) $11 \in L(n)$ if and only if $n \in \{3,5,11\}$.
(8) $l(C)$ is even if $LS(C) \neq \{0,1,\ldots,n-1\}$.
(9) Assume that n is odd. Then $l(C) \ge n+6$ if C is an odd non-fundamental cycle.

4 Some basic subgraphs in CCC_n

In [5], the authors introduced two basic subgraphs, X_n and Y_n in CCC_n. See Figure 1 for X_5 and Figure 2 for Y_5. And they have the following two lemmas.

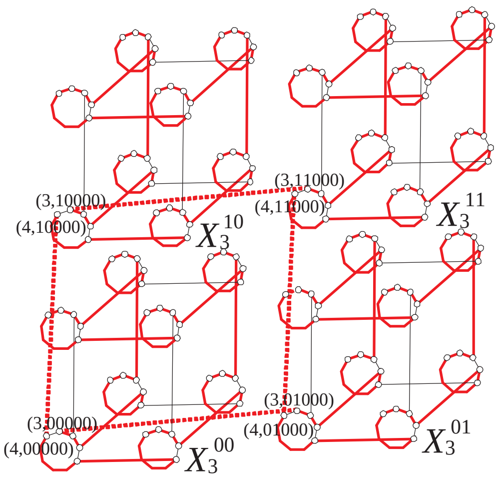

Figure 1: X_5

Lemma 4.1 *[5] Let n be an odd integer with $n \ge 3$. X_n forms a Hamiltonian cycle of CCC_n. Moreover, $((n-1,\mathbf{x}),(0,\mathbf{x})) \in E(X_n)$ for every n-bit binary string $\mathbf{x}$.*

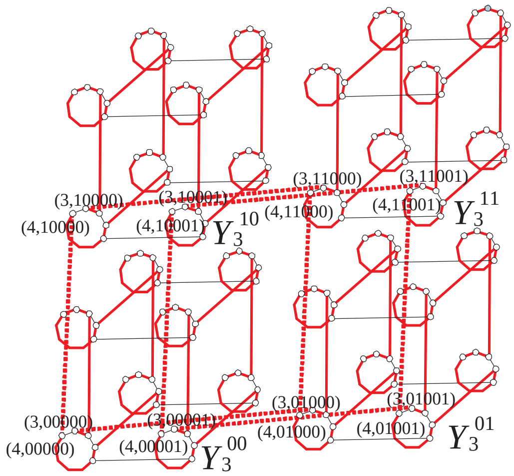

Figure 2: Y_5

Lemma 4.2 *[5] Let n be an odd integer with $n \ge 3$. Y_n forms two disjoint cycles such that one cycle spans the set $\{(l,\mathbf{x}) \mid \mathbf{x}$ is an n-bit binary string such that $(\mathbf{x})_0 = 0\}$ and the other cycle spans the set $\{(l,\mathbf{x}) \mid \mathbf{x}$ is an n-bit binary string such that $(\mathbf{x})_0 = 1\}$. Moreover, $((n-1,\mathbf{x}),(0,\mathbf{x})) \in E(Y_n)$ for every n-binary string $\mathbf{x}$.*

5 Construction scheme 1

In this section, we recursively construct a cycle $Z_{n,k}$ of length $n2^n - k$ in CCC_n if n is odd with $n \ge 3$ and k is odd with $1 \le k \le n-2$.

Let $Z_{3,1}$ be the subgraph of CCC_3 shown in Figure 3. It is easy to observe that $Z_{3,1}$ forms a cycle of CCC_3 such that vertex $(0,\mathbf{0}^3)$ is not in $Z_{3,1}$. Moreover, $((2,\mathbf{x}),(0,\mathbf{x})) \in E(Z_{3,1})$ for every non-zero 3-bit binary string $\mathbf{x}$.

Let n be an odd integer with $n \ge 5$. Assume that there exists a cycle $Z_{n-2,r}$ of length $(n-2)2^{n-2}-r$ in CCC_{n-2} with $1 \le r \le (n-2)-2$ such that $((n-3,\mathbf{x}),(0,\mathbf{x})) \in E(Z_{n-2,r})$ for every non-zero $(n-2)$-bit binary string $\mathbf{x}$.

Case 1. $Z_{n,n-2}$. We construct a spanning subgraph of CCC_n with edge set $E(Z^{00}_{n-2,n-4}) \cup E(X^{01}_{n-2}) \cup E(X^{11}_{n-2}) \cup E(X^{10}_{n-2})$. It is easy to observe that $((n-1,\mathbf{x}),(0,\mathbf{x})) \in E(Z^{00}_{n-2,n-4}) \cup E(X^{01}_{n-2}) \cup E(X^{11}_{n-2}) \cup E(X^{10}_{n-2})$ for every non-

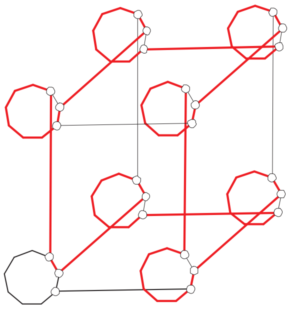

Figure 3: $Z_{3,1}$

zero n-bit binary string $\mathbf{x}$. We delete the edge set $\{((n-1, ab\mathbf{0}^{n-3}1), (n-2, ab\mathbf{0}^{n-3}1)) \mid ab \in \{00, 01, 10, 11\}\}$ and add the edge set $\{((n-2, 00\mathbf{0}^{n-3}1), (n-2, 01\mathbf{0}^{n-3}1)), ((n-2, 10\mathbf{0}^{n-3}1), (n-2, 11\mathbf{0}^{n-3}1)), ((n-1, 00\mathbf{0}^{n-3}1), (n-1, 10\mathbf{0}^{n-3}1)), ((n-1, 01\mathbf{0}^{n-3}1), (n-1, 11\mathbf{0}^{n-3}1))\}$. The resultant graph is $Z_{n,n-2}$. See Figure 4 for $Z_{5,3}$. The edges of $Z_{5,3}$ are shown using red lines and the added edges are shown using dotted lines.

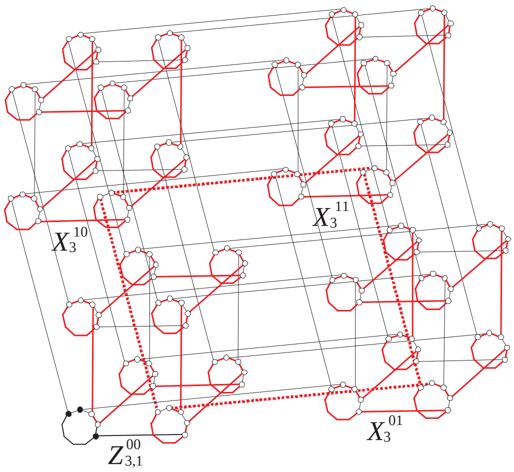

Figure 4: $Z_{5,3}$

Case 2. $Z_{n,k}$ where k is odd and $1 \leq k \leq n-4$. We construct a spanning subgraph of CCC_n with edge set $E(Z^{00}_{n-2,k}) \cup E(X^{01}_{n-2}) \cup E(X^{11}_{n-2}) \cup E(X^{10}_{n-2})$. It is easy to observe that $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(Z^{00}_{n-2,k}) \cup E(X^{01}_{n-2}) \cup E(X^{11}_{n-2}) \cup E(X^{10}_{n-2})$ for every non-zero n-bit binary string $\mathbf{x}$. We delete the edge set $\{((n-1, ab\mathbf{0}^{n-3}0), (n-2, ab\mathbf{0}^{n-3}0)), ((n-1, ab\mathbf{0}^{n-3}1), (n-2, ab\mathbf{0}^{n-3}1)) \mid ab \in \{00, 01, 10, 11\}\}$ and add the edge set $\{((n-1, 00\mathbf{0}^{n-2}), (n-2, 00\mathbf{0}^{n-2})), ((n-2, 00\mathbf{0}^{n-3}0), (n-2, 01\mathbf{0}^{n-3}0)), ((n-2, 00\mathbf{0}^{n-3}1), (n-2, 01\mathbf{0}^{n-3}1)), ((n-2, 10\mathbf{0}^{n-3}0), (n-2, 11\mathbf{0}^{n-3}0)), ((n-2, 10\mathbf{0}^{n-3}1), (n-2, 11\mathbf{0}^{n-3}1)), ((n-1, 00\mathbf{0}^{n-3}0), (n-1, 10\mathbf{0}^{n-3}0)), ((n-1, 00\mathbf{0}^{n-3}1), (n-1, 10\mathbf{0}^{n-3}1)), ((n-1, 01\mathbf{0}^{n-3}0), (n-1, 11\mathbf{0}^{n-3}0)), ((n-1, 01\mathbf{0}^{n-3}1), (n-1, 11\mathbf{0}^{n-3}1))\}$. The resultant graph is $Z_{n,k}$. See Figure 5 for $Z_{5,1}$. The edges of $Z_{5,1}$ are shown using red lines and the added edges are shown using dotted lines.

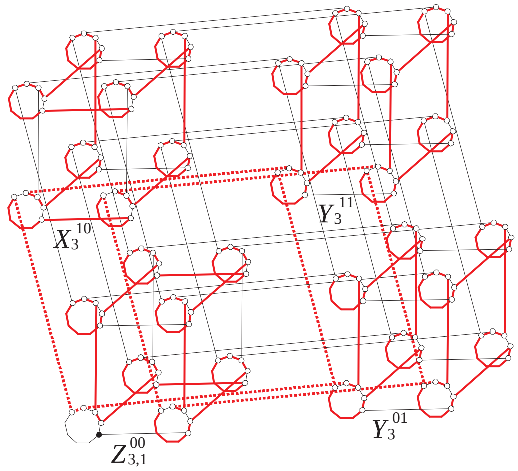

Figure 5: $Z_{5,1}$

Obviously, we have the following lemma.

Lemma 5.1 *Assume that n is odd with $n \geq 3$ and k is odd with $1 \leq k \leq n-2$. $Z_{n,k}$ is a cycle of length $n2^n - k$ in CCC_n such that $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(Z_{n,n-2})$ for every non-zero n-bit binary string $\mathbf{x}$.*

6 Construction scheme 2

In this section, we present a construction scheme for a cycle $Z_{n,2}$ of length $n2^n - 2$ in CCC_n with n is odd and $n \geq 5$.

By Lemma 5.1, there exists a cycle $Z_{n-2,1}$ of length $(n-2)2^{n-2} - 1$ in CCC_{n-2}. We construct a spanning subgraph of $CCC_n - \{(0, 00\mathbf{0}^{n-2})(0, 11\mathbf{0}^{n-2})\}$ with edge set $E(Z^{00}_{n-2,1}) \cup E(X^{01}_{n-2}) \cup E(Z^{11}_{n-2,1}) \cup E(Y^{10}_{n-2})$.

We delete the edge set $\{((n-1, ab\mathbf{0}^{n-3}0), (n-2, ab\mathbf{0}^{n-3}0)), ((n-1, ab\mathbf{0}^{n-3}1), (n-2, ab\mathbf{0}^{n-3}1)) \mid ab \in \{00, 01, 10, 11\}\}$ and add the edge set $\{((n-1,\ 00\mathbf{0}^{n-2}),\ (n-2,\ 00\mathbf{0}^{n-2})),\ ((n-1,\ 11\mathbf{0}^{n-2}),\ (n-2,\ 11\mathbf{0}^{n-2})),\ ((n-2,\ 00\mathbf{0}^{n-3}0),\ (n-2,\ 01\mathbf{0}^{n-3}0)),\ ((n-2,\ 00\mathbf{0}^{n-3}1),\ (n-2,\ 01\mathbf{0}^{n-3}1)),\ ((n-2,\ 10\mathbf{0}^{n-3}0),\ (n-2,\ 11\mathbf{0}^{n-3}0)),\ ((n-2,\ 10\mathbf{0}^{n-3}1),\ (n-2,\ 11\mathbf{0}^{n-3}1)),\ ((n-1,\ 00\mathbf{0}^{n-3}0),\ (n-1,\ 10\mathbf{0}^{n-3}0)),\ ((n-1,\ 00\mathbf{0}^{n-3}1),\ (n-1,\ 10\mathbf{0}^{n-3}1)),\ ((n-1,\ 01\mathbf{0}^{n-3}0),\ (n-1,\ 11\mathbf{0}^{n-3}0)),\ ((n-1,\ 01\mathbf{0}^{n-3}1),\ (n-1,\ 11\mathbf{0}^{n-3}1))\}$. See Figure 6 for $Z_{5,2}$. The edges of $Z_{5,2}$ are shown using red lines and the added edges are shown using dotted lines. Obviously, $Z_{n,2}$ forms a cycle of CCC_n of length $n2^n - 2$ such that vertices $(0, 00\mathbf{0}^{n-2})$ and $(0, 11\mathbf{0}^{n-2})$ are not in $Z_{n,2}$, and we have the following lemma.

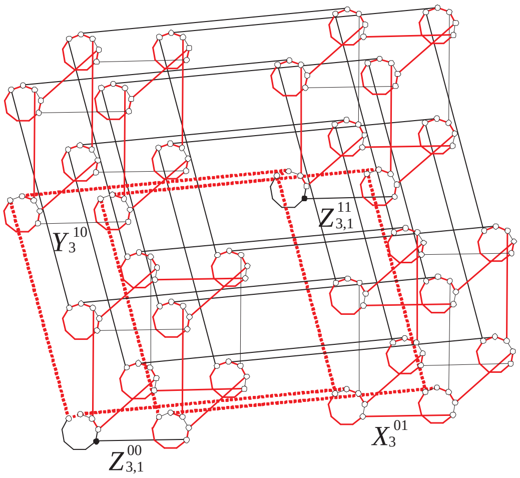

Figure 6: $Z_{5,2}$

Lemma 6.1 *Assume that n is an odd integer with $n \geq 5$. $Z_{n,2}$ is a cycle of length $n2^n - 2$ in CCC_n. Moreover, $Z_{n,2}$ can be written as $\langle(1, \mathbf{0}^{n-3}100), L, (2, \mathbf{0}^{n-3}100), (2, \mathbf{0}^{n-3}000), \ldots, (1, \mathbf{0}^{n-3}100)\rangle$ where L is the path $\langle(1, \mathbf{0}^{n-3}100), (0, \mathbf{0}^{n-3}100), \ldots, (2, \mathbf{0}^{n-3}100)\rangle$.*

7 Construction scheme 3

In this section, we present a construction scheme for a cycle $Z_{n,n}$ of length $n2^n - n$ in CCC_n with n is odd and $n \geq 5$.

To construct the cycle $Z_{n,n}$, we replace the subpath $\langle(1, \mathbf{0}^{n-3}100), L, (2, \mathbf{0}^{n-3}100)\rangle$ of $Z_{n,2}$ with the edge $((1, \mathbf{0}^{n-3}100), (2, \mathbf{0}^{n-3}100))$. Obviously, $\langle(1, \mathbf{0}^{n-3}100), (2, \mathbf{0}^{n-3}100), (2, \mathbf{0}^{n-3}000), \ldots, (1, \mathbf{0}^{n-3}100)\rangle$ is cycle of CCC_n of length $n2^n - n$. See Figure 7 for $Z_{5,5}$. Obviously, we have the following lemma.

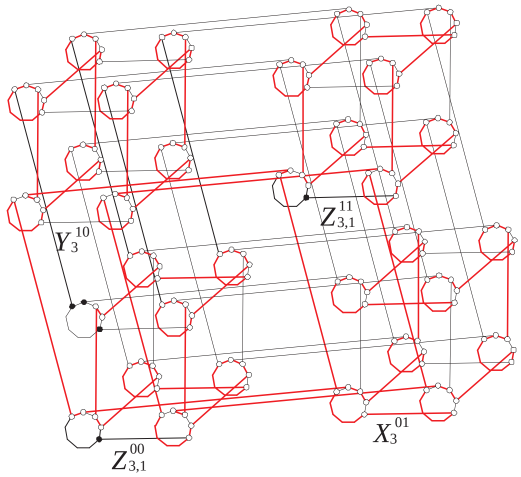

Figure 7: $Z_{5,5}$

Lemma 7.1 *Assume that n is an odd integer with $n \geq 5$. $Z_{n,n}$ is a cycle of length $n2^n - n$ in CCC_n.*

8 Proof of Theorem 1.3

With Theorem 1.2 and Lemma 3.1, we only need to find cycles of length $n2^n - k$ in CCC_n for the following case: $1 \leq k \leq n$ for n is odd with $n \geq 5$ and k is odd.

We consider $1 \leq k \leq n$ for n is odd with $n \geq 5$ and k is odd. Suppose that $1 \leq k \leq n-2$. By Lemma 5.1, $Z_{n,k}$ is a cycle of length $n2^n - k$ in CCC_n for any odd integer k with $1 \leq k \leq n-2$. Suppose $k = n$. By Lemma 7.1, $Z_{n,n}$ is a cycle of length $n2^n - n$ in CCC_n. The theorem is proved.

References

[1] F. Annexstein, M. Baumslag, A.L. Rosenberg, Group action graphs and parallel architectures, SIAM Journal of Computers 19 (1990) 544–569.

[2] J. Bruck, R. Cypher, and C.T. Ho, On the construction of fault-tolerant cube-connected cycles networks, Journal of Parallel and Distributed Computing, 25 (1995) 98–106.

[3] I. Friš, I. Havel, and P. Liebl, The diameter of the cube-connected cycles, Information Processing Letters, 61 (1997) 157–160.

[4] A. Germa, M.C. Heydemann, D. Sotteau, Cycles in the cube-connected cycles graph, Discrete Applied Mathematics 83 (1998) 135–155.

[5] T.Y. Ho, Z.M. Wei, C.W. Tsay, and L.H. Hsu, Fault-Tolerant Hamiltonicity of Cube-Connected Cycles Graphs, submitted to Applied Mathematics Letters, 2011.

[6] L.H. Hsu, C.K. Lin, Graph Theory and Interconnection Networks, CRC Press, 2008.

[7] F.P. Preparata, V. Vuillemin, The cube-connected cycles: a versatile network for parallel computation, commun. of the ACM 24 (1981) 300–309.

[8] R. Stong, On hamiltonian cycles in Cayley graphs of wreath products, Discrete Mathematics 85 (1987) 75–80.